

Math 514, F13
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $S(t)$, $t \geq 0$ be a geometric Brownian motion process with drift parameter $\mu = 0.1$ and volatility parameter $\sigma = 0.4$. Find

(a) $P(S(1) > S(0))$.

$$\begin{aligned} &= P\left(\frac{S(1)}{S(0)} > 1\right) = P\left(\log\left(\frac{S(1)}{S(0)}\right) > 0\right) = P(X(1) > 0) \\ &= P\left(Z > \frac{0 - \mu \cdot 1}{\sqrt{\sigma^2 \cdot 1}}\right) = P\left(Z > \frac{0 - 0.1 \cdot 1}{\sqrt{0.4^2}}\right) \\ &= P\left(Z > -\frac{1}{4}\right) = P\left(Z < \frac{1}{4}\right) \\ &= \Phi\left(\frac{1}{4}\right) \approx 0.5987 \end{aligned}$$

(b) $P(S(3) < S(1) > S(0))$.

$$\begin{aligned} P(S(3) < S(1)) &= P\left(\frac{S(3)}{S(1)} < 1\right) = P\left(\log\left(\frac{S(3)}{S(1)}\right) < 0\right) = P(X(2) < 0) \\ &= P\left(Z < \frac{0 - \mu \cdot 2}{\sqrt{\sigma^2 \cdot 2}}\right) = P\left(Z < \frac{0 - 0.1 \cdot 2}{\sqrt{0.4^2 \cdot 2}}\right) = P\left(Z < -\frac{\sqrt{2}}{4}\right) \\ &= 1 - \Phi\left(\frac{\sqrt{2}}{4}\right) \approx 0.3618 \end{aligned}$$

$$\begin{aligned} P(S(3) < S(1) > S(0)) &= P(S(3) < S(1), S(1) > S(0)) \\ &= P(S(3) < S(1)) \cdot P(S(1) > S(0)) \\ &\approx 0.3618 \times 0.4013 \approx 0.2166 \end{aligned}$$

(c) $E[S(3)]$ and $\text{Var}(S(3))$ given $S(0) = 2$.

$$\begin{aligned} S(3) &= S(0) e^{X(3)} = 2 e^{X(3)} \text{ where } E[X(3)] = \mu \cdot 3 = 0.3, \text{Var}(X(3)) = \sigma^2 \cdot 3 = 0.48. \\ E[S(3)] &= 2 E[e^{X(3)}] = 2 e^{E[X(3)] + \frac{1}{2} \text{Var}(X(3))} = 2 e^{0.3 + \frac{0.48}{2}} = 2 e^{0.54} \approx 3.432 \\ \text{Var}(S(3)) &= E[S(3)^2] - E[S(3)]^2 = E[(2e^{X(3)})^2] - E[2e^{X(3)}]^2 \\ &= E[4e^{2X(3)}] - (2E[e^{X(3)}])^2 \\ &= 4E[e^{2X(3)}] - 4(E[e^{X(3)}])^2 \\ &= 4 \cdot (e^{E[2X(3)] + \frac{1}{2} \text{Var}(2X(3))}) - 4(e^{0.54})^2 \\ &= 4(e^{0.6 + 0.96} - e^{1.08}) \\ &= 4e^{1.08}(e^{0.48} - 1) \approx 7.2566 \end{aligned}$$